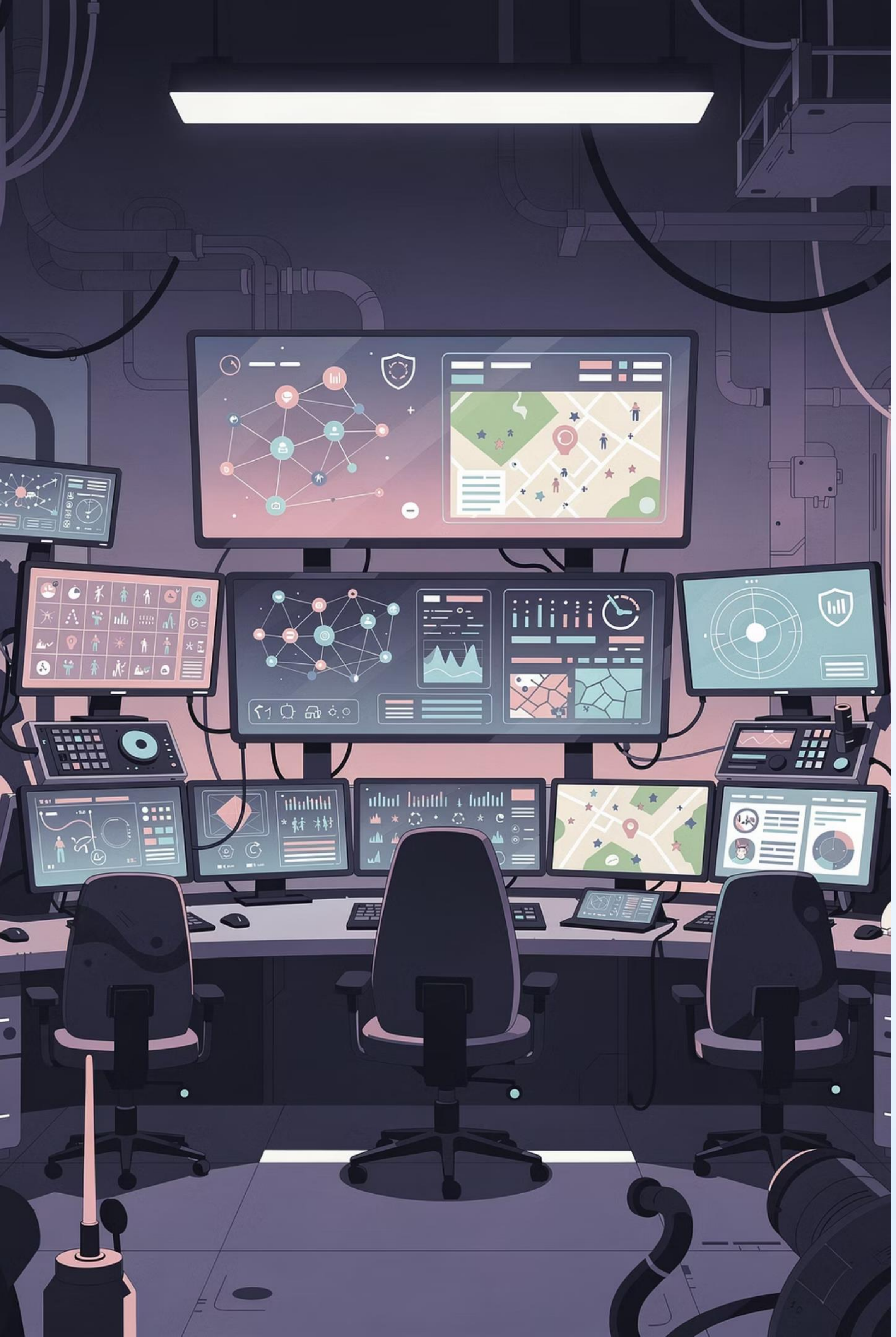


MODULE 05

Asset Inventory and Risk Register for OT Systems

Course: Cybersecurity for Embedded & OT Systems

Focus: Visibility → Criticality → Risk control (safety-first)

MODULE 04 RECAP

Recap — Module 04: Industrial Communication Protocols (Security View)

What We Learned in Module 04

- Industrial communication protocols act as the nervous system of OT environments
- OT protocols are designed for determinism, simplicity, and reliability, not security-first
- Communication enables data acquisition, control commands, alarms, and coordination
- Predictable and repetitive traffic patterns make abnormal behavior easier to detect
- Understanding how systems communicate is essential for safe industrial monitoring

Key Protocol Concepts Covered

Modbus

Uses a polling-based master–slave model with cyclic communication

MQTT

Follows a publish/subscribe model with brokers and topic-based messaging

OPC UA

Service-oriented protocol with structured data and integrated security concepts

Each protocol has different visibility and governance requirements. Misconfiguration and poor access control can introduce risks even in modern protocols.

Security & Monitoring Foundations

- OT messages contain valuable metadata such as source, destination, timestamp, and function codes
- Timing indicators like heartbeats, cycle time, and jitter reveal system health
- Logging supports troubleshooting, anomaly detection, and accountability
- Passive monitoring is preferred to avoid disrupting real-time operations
- Protocol-level risk awareness supports layered, safety-first defense strategies

Secure communication requires knowing who talks, when they talk, and what is normal.
The next module builds on this awareness to strengthen asset visibility and risk management.

CRITICAL FOUNDATION

Why This Module Matters

Operational Technology incidents often happen not because of sophisticated attacks, but because teams lack fundamental knowledge about their own environment. Without clear visibility, organizations cannot effectively protect their critical infrastructure or respond to emerging threats.

Unknown Assets

What devices exist in the environment

Location Mystery

Where critical systems are physically located

Ownership Gaps

Who is responsible for each asset

Criticality Blind Spots

Which systems are essential for safe operation

Asset visibility directly prevents accidental downtime, unsafe changes, and slow incident response. The fundamental principle: You cannot protect what you do not know.

What Is "Asset Visibility"?

Asset visibility means maintaining a clear, continuously updated view of your operational technology environment. It goes beyond simply knowing what devices exist—it encompasses understanding their configuration, connections, responsibilities, and operational importance.

Physical Devices

PLCs, sensors, drives, HMIs, and controllers

Software & Firmware

Version tracking and update status

Network Location

Zones, segments, and communication paths

Ownership

Assigned responsibilities and accountability

Criticality Rating

Operational importance and risk level

> **Outcome:** Better protection strategies, safer maintenance procedures, and faster troubleshooting when incidents occur.

Consequences of Poor Visibility

When organizations lack comprehensive asset visibility, they face cascading risks that compound over time. Each visibility gap creates vulnerabilities that can lead to safety incidents, operational disruptions, or security breaches.

Unknown Devices

Unmanaged systems operate outside security controls, creating blind spots in monitoring and protection strategies. Shadow IT in OT environments is particularly dangerous.

Unclear Ownership

When no one knows who is responsible for an asset, incident response slows dramatically. Critical decisions get delayed while teams identify the right stakeholders.

Missing Version Data

Without firmware and software version tracking, updates become dangerous. Teams cannot assess compatibility or safety impacts before making changes.

Network Path Confusion

Unclear network topology makes troubleshooting painfully slow. Engineers waste precious time tracing connections during critical incidents.

No Criticality Ranking

Without prioritization, organizations underprotect their most important assets while over-investing in low-risk systems. Resources get misallocated.

COMPREHENSIVE VIEW

Types of Assets in OT (Not Only Devices)

Effective OT asset management requires looking beyond physical hardware. A comprehensive inventory captures all elements that contribute to operational capability and safety.

Hardware Assets

Programmable logic controllers, sensors, relays, variable frequency drives, actuators, and field devices that directly control physical processes.

Software Assets

Firmware versions, PLC logic programs, HMI applications, SCADA configurations, and embedded operating systems that enable device functionality.

Network Assets

Industrial switches, network cables, IP links, wireless access points, and zone boundaries that enable communication between systems.

Data Assets

Critical setpoints, alarm limits, historical logs, configuration backups, and operational parameters that define system behavior.

Documentation Assets

Electrical drawings, system manuals, standard operating procedures, network diagrams, and vendor documentation essential for safe operations.

Critical insight: Documentation and configuration files are assets that require the same protection and version control as physical devices.

Bridge from Module 04: Protocol Visibility

Protocol visibility from Module 04 connects directly to asset inventory and risk management. By monitoring Modbus and other industrial protocols, teams gain critical intelligence about their OT environment.

What to Observe in Modbus Traffic

Defensive monitoring focuses on behavior patterns that reveal asset changes or anomalies:

- **New or unknown devices responding** to Modbus queries
- **Unexpected function codes** that deviate from baseline
- **Changes in polling frequency** from normal patterns
- **Write operations** where only read operations are expected

Monitoring Focus

The goal is understanding normal operations to detect deviations quickly:

- **Behavior patterns**

  Establish baselines for normal communication

- **Timing analysis**

  Track frequency and duration of operations

- **Access validation**

  Verify authorized communication paths

> This visibility connects directly to identifying assets and risks in Module 05. What you observe in protocol traffic helps validate and enhance your asset inventory.

Building an OT Asset Inventory

An OT asset inventory is a structured, living record of all assets within a defined operational scope. Whether you're documenting a single pump cell, a control panel, or an entire production line section, the approach remains consistent and methodical.

Define Boundary

Clearly establish what is inside the inventory scope. Be specific about physical and logical boundaries to avoid confusion.

Identify Assets

Catalog all devices and supporting assets within the boundary, including hardware, software, network elements, and documentation.

Record Key Fields

Document essential attributes for each asset using a consistent template that captures ownership, location, version, and criticality.

Assign Owner

Designate a specific person or team responsible for each asset to ensure accountability during maintenance and incidents.

Validate & Update

Review the inventory regularly and update it whenever changes occur. An outdated inventory is nearly as dangerous as no inventory.

What Fields Should Your Inventory Contain?

An effective OT asset inventory balances comprehensiveness with usability. Too few fields leave gaps in knowledge; too many create maintenance burden. Focus on fields that directly support operational safety and security decisions.

1

Asset ID

Unique identifier for tracking and reference

2

Asset Name & Type

Descriptive name and device category

3

Physical Location

Specific room, panel, or equipment area

4

Owner/Team

Responsible person or department

5

Firmware/Version

Current software and firmware versions

6

Network Zone & IP

Logical zone and network address

7

Criticality

High, Medium, or Low operational importance

Goal: Make the inventory immediately useful during maintenance activities and incident response situations. The list should answer questions quickly without requiring additional research.

☝ PRIORITY FIELDS

Three Attributes That Matter Most

While every inventory field adds value, three attributes are particularly crucial for transforming a basic asset list into an actionable security tool. These fields drive accountability, enable safe change management, and support network security architecture.

1. Owner

Enables: Accountability

Clear ownership eliminates confusion during incidents. When something goes wrong, teams immediately know who to contact. During maintenance windows, ownership defines approval authority.

Without ownership assignment, response times increase dramatically as teams waste time identifying stakeholders.

2. Firmware/Version

Enables: Change Tracking & Safety Impact

Version data allows teams to assess update compatibility and safety implications before making changes. It supports vulnerability management by identifying which systems need patches.

Version tracking also helps diagnose issues by correlating problems with software changes.

3. Network Location/Zone

Enables: Exposure & Communication Boundaries

Network zone information reveals which systems can communicate with each other. It defines security boundaries and helps teams understand attack paths.

Zone data is critical for segmentation strategies and firewall rule management.

Result: Less confusion, faster safe decision-making, and more effective risk management across the OT environment.

Identifying "Crown Jewels"

Crown jewel assets are those whose failure or compromise causes the highest impact to safety, operations, or business continuity.

Identifying these assets allows organizations to focus protection efforts where they matter most.

How to Identify Crown Jewels

Direct Process Control

Assets that directly command physical processes

Safety-Critical Function

Systems that prevent hazardous conditions

Single Point of Failure

No redundancy or backup capability

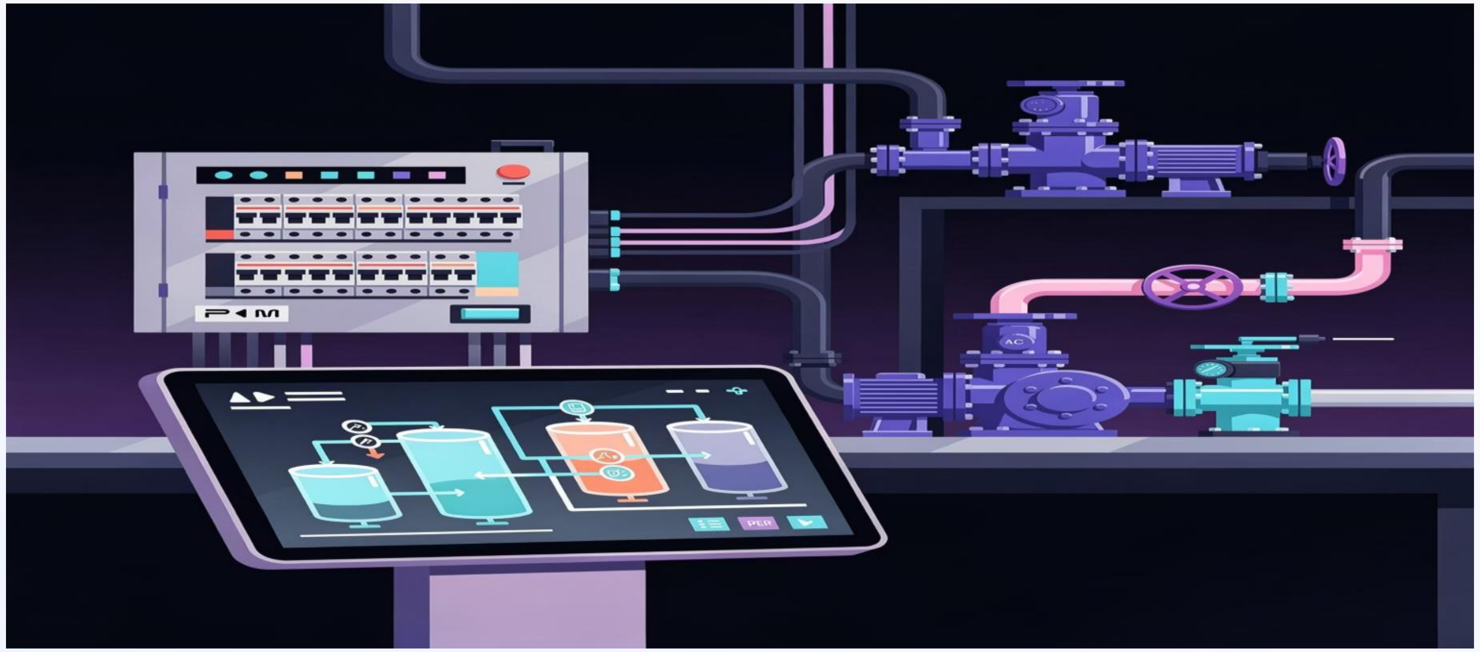
Replacement Difficulty

Long lead times or specialized equipment

Business/Regulatory Impact

High financial or compliance consequences

Example Crown Jewels: Pump Cell



PLC Controller

Primary logic controller for pump operations

Safety Interlock

Emergency shutdown and protective systems

HMI Setpoint Screen

Operator interface for critical parameters

Network Switch

Communication hub for control network

Alarm Configuration

Safety and operational alarm settings

Introduction to OT Risk Management

Risk management in OT environments requires a practical, safety-first approach that balances thoroughness with operational reality. Unlike IT risk management, OT risk assessment must account for physical consequences and operational continuity.

What We Aim to Reduce

- Unsafe operations that threaten people or environment
- Unplanned downtime that disrupts production
- Uncontrolled changes that introduce instability

Risk Components

Likelihood: How possible is the risk scenario?

Impact: How severe would the consequences be?

Risk level emerges from the combination of these factors.

OT Risk Management Must Be:

Simple

Easy to understand and apply

Practical

Actionable in real operations

Safety-First

Prioritizes physical safety

Qualitative Risk: Likelihood & Impact

OT environments benefit from qualitative risk assessment that provides clear prioritization without complex quantitative calculations. Simple likelihood and impact scales enable rapid risk evaluation during operational decision-making.

Low Likelihood

Unlikely to occur under normal operating conditions. Requires multiple control failures or unusual circumstances to manifest.

Medium Likelihood

Possible during typical operations, especially during transitions or elevated activity periods. May occur without additional controls.

High Likelihood

Likely to occur without intervention or improved controls. May already be occurring or narrowly avoided through informal mitigations.

Low Impact

Minor inconvenience or temporary degradation. Minimal effect on safety, production, or business operations. Quick recovery possible.

Medium Impact

Temporary production impact or reduced operational capacity. Requires coordinated response but no safety consequences. Recovery within hours.

High Impact

Safety risk to personnel or environment, major downtime, or significant business impact. Extended recovery time or permanent consequences.

Purpose: Enable rapid risk prioritization without lengthy analysis. Teams can quickly assess and act on risks during change windows and incident response.

Simple OT Risk Register (What It Contains)

A risk register is a living document that tracks identified risks, their assessments, and mitigation strategies. It serves as a working tool for operations teams, not a compliance checkbox.

Risk ID
Unique tracking number

Affected Asset
Which system or component

Risk Description
Clear scenario statement

Likelihood & Impact
Qualitative assessment

Risk Level
Combined priority rating

Mitigation Action
Specific control or response

Common OT Risk Register Examples

Risk Scenario	Asset Type	Risk Level	Mitigation Action
Unauthorized setpoint change	HMI	High	Implement role-based access control
Single switch failure	Network switch	High	Add redundant switch connection
Shared HMI login credentials	Operator stations	Medium	Deploy unique user accounts
Outdated firmware with vulnerabilities	PLC	Medium	Schedule planned update window
Sensor drift affecting control	Process sensors	Medium	Establish periodic calibration checks

Link Risk Management with Change Control

Risk management and change control are inseparable in OT environments. Every proposed change should trigger a risk assessment to ensure safe implementation. This integration prevents unintended consequences and maintains operational stability.

Firmware Update

Assess compatibility, safety impact, and rollback procedures before deploying new firmware versions.

Network Change

Review communication risk, zone boundaries, and access implications before modifying network topology.

Access Changes

Re-evaluate insider risk and segregation of duties when modifying user permissions or credentials.

Hardware Replacement

Update inventory, verify compatibility, and document configuration changes for new equipment.

Vendor Access

Update risk register with third-party access details, duration, and monitoring requirements.

> **Key idea:** Change control keeps operations stable and safe. When risk management is integrated into change processes, teams catch potential issues before they become incidents.

MODULE SUMMARY

Module 05 Key Takeaways

This module established the foundation for effective OT security through comprehensive asset visibility and practical risk management. These concepts enable informed decision-making about protection priorities and change control.

Asset visibility is the base of OT security

You cannot protect what you don't know. Comprehensive visibility enables effective protection strategies and rapid incident response.

OT assets extend beyond devices

Assets include hardware, firmware, configurations, network elements, and documentation—all requiring tracking and protection.

Inventory needs critical attributes

Owner, firmware/version, physical location, and network zone are essential fields that drive accountability and safe change management.

Crown jewels drive protection priority

Identifying high-impact assets allows organizations to focus resources where failure or compromise would cause the greatest harm.

Risk equals Likelihood times Impact

Qualitative assessment provides sufficient prioritization for OT environments without complex calculations.

Risk register plus change control equals safety

Integrating risk assessment into change control processes prevents unintended consequences and maintains operational stability.

MODULE 05 — Asset Inventory & Risk Register for OT Systems

Reflection & Class Discussion

CLASS DISCUSSION

- Why asset visibility is the foundation of OT cybersecurity
- Difference between hardware assets and supporting assets
- Why ownership and firmware details matter during incidents
- Meaning of crown jewels in an OT environment
- How risk management supports safety and availability

HOMEWORK & ASSIGNMENTS

1. List all asset types in a small OT setup (minimum 5 categories)
2. Create a simple asset inventory table for a pump control cell
3. Identify three crown jewel assets and justify their criticality
4. Perform a qualitative risk assessment (likelihood + impact) for two assets
5. Prepare a basic OT risk register with at least three risks

OPTIONAL CHALLENGE

Risk Mapping Exercise:

Choose one OT asset and document:

- Asset attributes (owner, firmware, network zone)
- One realistic risk
- Likelihood level
- Impact level
- One practical mitigation action

What's Next: Module 06 Preview

COMING NEXT

Segmentation, Access Control, and Monitoring

Having established comprehensive asset visibility and risk understanding, we now move into implementing protective controls. Module 06 focuses on practical security measures that reduce exposure and improve detection capabilities.

Apply controls around crown jewels

Implement layered defenses for your most critical assets

Reduce exposure using zones and boundaries

Leverage segmentation, conduits, and security zones

Improve monitoring using logs and visibility

Deploy safe monitoring that doesn't disrupt operations

Strengthen access governance

Control access to OT devices, tools, and interfaces

Thank you. We move from "knowing assets" to "protecting them" with practical controls that enhance both security and operational safety.